

It's a Price! It's a Value!

Economic Measurement Across Environments

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Abstract

Revaluing the same economic object across environments requires a measurement profile. Under stable mixed object–environment kernels, the local relational coordinates have dimension equal to the mixed interaction rank, and a reference valuation supplies the remaining level information. Retained source values then define a residual target map. In a bilinear system, the rank of this map gives the minimum number of linear certificates needed for exact recovery. Its covariance spectrum gives prediction loss and the marginal value of another record. The F-MAP application treats source price environments and basket-composition certificates as alternative fields that can be retained. With four source records, the implemented greedy target-directed rule reduces held-out MSE by 71.7 percent relative to dispersed selection. One-exchange refinement raises the reduction to 77.7 percent; its fixed-policy loss effect has a simultaneous 95-percent interval of $[0.0367, 0.0927]$. Across 499 public-unit-value re-extractions crossed with ten held-out node perturbations, the four-record exchange rule has a 75.9-percent median reduction and a central range of $[44.2, 90.1]$ percent. Two composition records complete the two residual dimensions and recover 57,600 held-out basket–target valuations to numerical precision. Treasury cash flows provide a second application using duration, convexity, and key-rate exposure records.

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1. Introduction

A consumption basket has an expenditure at each price vector, and a stream of promised payments has a present value at each discount curve. A record preserves one evaluation of the object. Revaluation requires a state for the object's valuations, a description of the target information left unresolved by retained records, and a rule for valuing an additional feasible record.

Record retention is an information policy. An institution decides which source valuations and object-side certificates will accompany an object into future environments. That decision depends on the target environments, the feasible menu of fields, available capacity, and the loss from valuation error. These primitives also determine the value of one additional field.

Let $m(x, z)$ be the value of object x in environment z , and let $\mathfrak{M}(x) = [z \mapsto m(x, z)]$ denote its measurement profile. Suppose the object kernel of the mixed derivative persists across environments and the environment kernel persists across objects. Theorem 2 then constructs a local relational quotient whose dimension equals the mixed derivative rank. Its coordinates describe how valuation responds to environmental change, while a value in one reference environment supplies the level. Corollary 3 uses this representation to bound the target-relevant information that can remain unresolved after observing a source record. The example $m(x, z) = x_1 z + x_2 z^2$ shows why kernel stability matters: every mixed derivative has rank one, but the active object direction rotates with z , so two coordinates are needed across environments.

Given retained source values $Y_O(x) = y$, define the residual transport set

$$(1) \quad \mathcal{R}_{O \rightarrow T}(y) = \{Y_T(x) : Y_O(x) = y\}.$$

This set contains all target valuations compatible with the retained record. Under the quotient conditions, (19) expresses its local dimension in the common state coordinates. The target-relative residual dimension counts source-preserving changes that still alter target values. A singleton set gives exact recovery. Under constant ranks, the dimension equals the minimum number of additional smooth coordinates required locally. Throughout the paper, transport refers to revaluing the same object across environments.

For exact bilinear valuation systems, Theorem 3 derives these quantities from one residual target map. Its rank determines exact recovery and the minimum linear certificate count. On a bounded exposure set it gives worst-case error, and under exposure moments and a target metric it gives average affine-prediction loss. The covariance spectrum records the loss remaining after each certificate, while a rank-one update gives the value of one additional scalar field. Environment records retain valuations at available price vectors

or discount curves. Object-side certificates retain composition or cash-flow exposure summaries. Their feasible menus differ, but both enter the same marginal-loss calculation.

The F-MAP application evaluates 57,600 held-out basket-target pairs. With four retained source values, greedy target-directed acquisition lowers MSE by 71.7 percent against dispersed acquisition. One-exchange refinement raises the reduction to 77.7 percent. A common-multiplier band across capacities places the four-record fixed-policy effect in $[0.0367, 0.0927]$, and its simultaneous RMSE upper bound is 4.74 basis points of benchmark expenditure. Combining public-unit-value policy reselection with held-out node perturbations gives a 75.9-percent median reduction and a $[44.2, 90.1]$ central range. The seven-group system has centered support rank six, so four source values leave two target-relevant directions; two calibration-based composition records complete those directions and recover the held-out array to numerical precision. Treasury cash flows provide a second application through duration, convexity, and key-rate exposure records.

Measurement and economic purpose. Schennach (2022) develops identification of latent distributions from systems of imperfect measurements. The recovery object here is the target valuation vector of the same economic object, conditional on retained source values. A source fiber may contain several objects even when its target image is a singleton. Almås et al. (2024) places the joint design of economic models and measurement tools on the measurement agenda. The recording decision studied here is specified by future valuations, permitted fields, capacity, and valuation loss. The residual frontier measures the information needed for that declared use.

Separability and aggregation. Functional decomposition and economic aggregation provide the ancestry of common measurement states (Leontief 1947a,b; Nataf 1948; Green 1964; Goldman and Uzawa 1964; Gorman 1968). Green’s synthesis separates within-function grouping based on within-group marginal-rate invariance from additive replacement of a subset based on pairwise marginal-rate restrictions. These results act on one domain and supply factorization antecedents. Theorem 2 starts from a cardinal map on the product $\mathcal{X} \times \mathcal{Z}$, imposes stability on both kernels of $D_x D_z m$, and matches both quotient dimensions to the mixed rank with a nonsingular induced mixed derivative. Its proof applies the classical factorization logic to each side and supplies the bilateral rank-matching step. Gorman (1959) studies group-price information sufficient for expenditure allocation under separability. Exact index numbers link index formulas to aggregators (Diewert 1976), and Divisia path independence is tied to aggregate existence (Gorman 1996). The valuation map uses declared economic units. The mixed-kernel condition acts on its cross-difference, yielding object and environment coordinates together with a baseline value. Constant-rank and Frobenius arguments supply the construction (Lee 2013); differential methods also have

an aggregation tradition (Chiappori and Ekeland 1999). Target sufficiency is then assessed on each declared source fiber. A multidimensional profile can have exact revaluation for selected targets, as the example in Section 2.2 shows.

Sufficiency and dimension reduction. Deterministic fiber factorization is a sufficiency argument within the comparison-of-experiments tradition (Blackwell 1953). Nonlinear sufficient dimension reduction develops response-relevant sub- σ -fields (Lee et al. 2013), and economic dimension reduction preserves restrictions on demand and its marginal effects (Hoderlein and Lewbel 2012). Active-subspace and vector-output methods use integrated derivative information to construct approximations (Constantine et al. 2014; Zahm et al. 2020); these operators can also register directions that rotate across environments. The exact question here concerns a common two-sided relational quotient whose dimension equals the interaction rank. Connecting that quotient to the residual source fiber gives the local information count used in the record frontier.

Goal-oriented design and compression. Target-oriented uncertainty reduction is developed by Attia et al. (2018); Zhong et al. (2026) treats nonlinear predictive quantities through Bayesian expected information gain. Chen et al. (2025) couples input and output reduction for a deterministic nonlinear map and derives bounds for goal-oriented sensor selection. Gaussian revaluation loss in the linear model equals the zero-noise limit of goal-oriented A-optimal loss; appendix C derives this correspondence. Projection, reduced-rank geometry, and optimal design supply the spectral and acquisition calculations (Fan 1949; Tso 1981; Elfving 1952; Kiefer and Wolfowitz 1960). Large-scale design computation and exchange search have established algorithmic literatures (Wu et al. 2023; Lau and Zhou 2022). The economic characterization links these calculations to the same-object source fiber, exact completion dimension, common-state representation, and admissible economic fields. Information-bottleneck methods trade representational rate against target relevance (Tishby et al. 1999). Capacity here counts specified smooth or linear scalar fields, with a shared dictionary and a feasible record menu. Encoding precision would enter as an additional constraint.

The measurement problem leads first to a common state and then to the residual-information and record frontier. F-MAP evaluates source-record choice, and Treasury cash flows show how object-side certificates repair missing valuation directions. Proofs, global quotient conditions, approximation inference, and empirical construction appear in the online appendix.

2. The Economic Measurement Problem

2.1. Objects, environments, and retained records

Let $\mathcal{X}$ be an economic-object space, $\mathcal{Z}$ an evaluation-environment space, and $m : \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{R}$ a declared measurement map. Expenditure $m(q, p) = p'q$ and present value $m(c, z) = d(z)'c$ evaluate a fixed quantity vector or dated cash-flow vector in different price or discount environments. Units, currency conversions, and base-period normalizations are fixed before environments are compared. Online appendix A states the corresponding normalization on equivalence classes.

DEFINITION 1 (Measurement profile). *The profile of x is $\mathfrak{M}(x) = [z \mapsto m(x, z)]$. Two objects are measurement equivalent, written $x \sim_M x'$, when their profiles agree. The profile image is $\mathcal{M} = \mathfrak{M}(\mathcal{X}/\sim_M)$.*

An institution retains a record from source environments $O = \{z_1, \dots, z_K\}$ and uses that record to value the same object at targets $T = \{z_1^*, \dots, z_J^*\}$. Define

$$(2) \quad Y_O(x) = (m(x, z_1), \dots, m(x, z_K))', \quad Y_T(x) = (m(x, z_1^*), \dots, m(x, z_J^*))'.$$

The capacity K counts retained scalar valuations. A policy chooses their environments from a feasible menu. The target family and a loss on its valuation coordinates specify the use of the record. Here *transport* means recovery of the same object's value across evaluation environments; its mathematical operation is factorization of evaluation maps.

DEFINITION 2 (Residual transport set). *For an attained source record $y \in Y_O(\mathcal{X})$, the residual transport set is*

$$(3) \quad \mathcal{R}_{O \rightarrow T}(y) = Y_T(Y_O^{-1}(y)) = \{Y_T(x) : Y_O(x) = y\}.$$

The source fiber $Y_O^{-1}(y)$ contains the objects compatible with the record, and its image under Y_T contains their compatible target valuations. The target vector can therefore be identified even when several objects remain in the source fiber. Sufficiency is target-specific: a record that recovers one target family may leave positive residual variation for another.

2.2. Target-specific revaluation

Consider three-good expenditure on an open set of positive quantity vectors at all positive price vectors. Its profile has dimension three. Retain expenditures at $p_1 = (1, 1, 1)'$ and $p_2 = (2, 1, 1)'$. The target $p_T = (3, 2, 2)' = p_1 + p_2$ is recovered as $p_T'q = p_1'q + p_2'q$. The source fiber permits offsetting changes in the second and third quantities along $(0, 1, -1)'$ while

both retained expenditures stay fixed. Thus a three-coordinate profile admits exact target revaluation from two source fields. The relevant sufficiency condition is constancy of the target on that fiber.

2.3. A composition record for revaluation

Consider two goods, source prices $p_O = (1, 1)'$, and target prices $p_T = (2, 1)'$. Baskets $q^A = (3/2, 1/2)'$ and $q^B = (1/2, 3/2)'$ both cost two at the source; they cost $7/2$ and $5/2$ at the target. Hence $\mathcal{R}_{O \rightarrow T}(2)$ contains both target values. The scalar composition record $C(q) = q_1 - q_2$ completes the comparison:

$$(4) \quad p'_T q = \frac{3}{2} p'_O q + \frac{1}{2} C(q).$$

An environment with prices $(2, 1)'$ and the object-side certificate C supply the same missing valuation direction through different permitted measurements. The choice between them depends on the institution's measurement menu and available object-side inputs.

2.4. Timing and the evaluation oracle

At the recording date, a source policy selects O using a calibration sample and a declared target specification. The institution retains $Y_O(x)$ and, when permitted, a certificate $C(x)$ formed from a declared pre-target input $P(x)$. At the revaluation date, the decoder receives the retained record and the target environment. Reporting, survey, and archival systems motivate the capacity constraint: future valuation uses the records that the system carries forward.

An operational record contains the source valuations and one scalar field for each certificate. A shared dictionary supplies the source environments, certificate weights, units, and decoder calibration. Quantity projections become composition-index fields; cash-flow projections become retained exposure summaries. Capacity counts scalar fields per object, while the recorded population shares the dictionary. The measurement choice concerns the information carried by each additional field.

The empirical laboratory observes quantities and prices to construct the accounting value for every held-out pair. These primitive data serve as an evaluation oracle. Policy predictions use only the retained source values, the chosen certificates, and calibration moments. The laboratory therefore evaluates the consequences of a reporting or retention rule. Its intervention is the information passed to the future decoder. Monetary collection costs and behavioral responses are additional application primitives.

The F-MAP application maps this timing to a data-publication chain. Commercial scanner inputs enter an upstream construction process, ERS publishes aggregate dollar,

gram, unit-value, and price-index fields, and a downstream user carries selected economic fields into later revaluation. The empirical policy treats published area-month environments as a feasible counterfactual reporting menu. USDA operating costs and current field-retention choices lie outside the reported effect. This separation places the institutional primitive in the declared reporting capacity and leaves its monetary cost available for an application-specific objective.

The residual set $\mathcal{R}_{O \rightarrow T}$ is used at three levels. If it is a singleton, recovery is exact. Under constant ranks, the dimension of its local target variation counts the additional smooth information required for recovery. A declared object set or exposure distribution, together with a target loss, turns the same variation into a valuation-loss frontier. Adding a source environment or certificate contracts this residual information.

3. Representation of the Measurement Profile

3.1. The scalar case

For each environment let $I_z = m(\mathcal{X}, z)$ be its attained range. An object-independent transport is a map $\tau_{z \rightarrow z'} : I_z \rightarrow I_{z'}$ satisfying

$$(5) \quad m(x, z') = \tau_{z \rightarrow z'}(m(x, z)) \quad \text{for every } x.$$

THEOREM 1 (Scalar transport representation). *On a nonempty object region, exact transports exist for every ordered environment pair if and only if there are a scalar state $L : \mathcal{X} \rightarrow \mathbb{R}$ and injective coordinates $\phi_z : L(\mathcal{X}) \rightarrow I_z$ such that*

$$(6) \quad m(x, z) = \phi_z(L(x)).$$

The transports are bijections on attained ranges and obey

$$(7) \quad \tau_{z' \rightarrow z''} \circ \tau_{z \rightarrow z'} = \tau_{z \rightarrow z''}.$$

Two such representations with strictly increasing coordinates have latent states related by a strictly increasing bijection.

To construct the representation, fix z_0 and take $L = m(\cdot, z_0)$. Exact reverse transports make each coordinate injective. This is the scalar starting case; online appendix A gives the proof and the refinements for attained ranges.

COROLLARY 1 (Ranking reversal). *If two objects exchange their strict ordering between two environments, the class in (6) with all coordinates strictly increasing is empty.*

Path independence here composes exact coordinate changes for a fixed object. In the index-number tradition, path independence of a Divisia integral follows from economic conditions supporting an aggregate (Gorman 1996). The source–target problem in Section 2 allows a vector profile and evaluates sufficiency separately for each declared target family.

3.2. Stable interaction kernels

A measurement state $s(x)$ generates the profile when $m(x, z) = g(s(x), z)$ for every environment. When the profile image is a smooth manifold, write $r_M = \dim \mathcal{M}$. The chain rule gives a lower bound $k \geq r_M$ for a smooth k -coordinate state; a chart of an embedded constant-rank profile image attains it locally. Online appendix A gives the function-space and evaluation assumptions.

Fix a reference pair (x_0, z_0) and remove the additive object and environment terms:

$$(8) \quad \Delta_0 m(x, z) = m(x, z) - m(x, z_0) - m(x_0, z) + m(x_0, z_0).$$

Its mixed derivative is $J_M(x, z) = D_x D_z m(x, z)$. A common relational quotient with dimension equal to the mixed rank requires the annihilated directions to persist as the opposing argument varies.

Economically, an object direction in the left kernel preserves the marginal response of valuation to every environmental change. Kernel stability requires the same response-irrelevant directions at each environment. The right-kernel condition imposes the corresponding requirement on environmental directions across objects. These restrictions concern the interaction component of value. An object direction can preserve that interaction and still change the reference valuation; Corollary 2 accounts for this level information. The map m and its units are fixed throughout the calculation.

ASSUMPTION 1 (Smooth interaction and stable kernels). *The measurement map is C^3 on a product of C^3 manifold neighborhoods $N_X \times N_Z$. Its left kernel $K_X(x) = \ker_L J_M(x, z)$ is independent of z , and its right kernel $K_Z(z) = \ker_R J_M(x, z)$ is independent of x . These kernels are smooth distributions of constant codimension r .*

THEOREM 2 (Local relational quotient). *Under assumption 1, K_X and K_Z are involutive. Locally there are r -coordinate submersions $u(x), v(z)$ with $\ker Du = K_X$, $\ker Dv = K_Z$, and*

$$(9) \quad m(x, z) = a(x) + b(z) + F(u(x), v(z)),$$

$$(10) \quad \Delta_0 m(x, z) = F_0(u(x), v(z)), \quad F_0(u, v_0) = F_0(u_0, v) = 0.$$

The induced mixed derivative $D_u D_v F_0$ has rank r . Conversely, such a factorization through submersions with nonsingular induced mixed derivative has the stated stable kernels.

The theorem produces interaction coordinates with matched ranks: each quotient has r coordinates, and the induced mixed derivative also has rank r . For fixed (z, η) , fields in K_X annihilate $D_z m[\eta]$, as do their brackets, so Frobenius provides quotient charts. Equivalently, r independent components of $D_z m(x, z_0)$ form local first integrals with kernel K_X ; the environment quotient follows symmetrically. The construction uses constant-rank and functional-factorization arguments (Lee 2013). Online appendix B gives the details and the global descent conditions.

COROLLARY 2 (Relational and full profile dimension). *Under the profile-manifold regularity in online appendix A, the relational profile has differential rank r . The full profile is locally generated by*

$$(11) \quad s_{z_0}(x) = (m(x, z_0), u(x)), \quad r \leq r_M \leq r + 1.$$

At regular points $r_M = r$ exactly when $m(\cdot, z_0)$ locally factors through u .

The full state expresses both source and target maps in the quotient coordinates. Retaining the reference value $m(x, z_0)$ fixes the level coordinate. Residual revaluation then depends on the interaction coordinates left unresolved by the remaining source records. The information frontier below measures precisely this unresolved part.

3.3. Rotation across environments

PROPOSITION 1 (Rotating-kernel example). *For*

$$(12) \quad m(x, z) = x_1 z + x_2 z^2, \quad J_M(x, z) = \begin{pmatrix} 1 \\ 2z \end{pmatrix},$$

the pointwise mixed rank is one, while the relational profile has differential rank two. The left kernel $\{(\xi_1, \xi_2) : \xi_1 + 2z\xi_2 = 0\}$ varies with z . On a product neighborhood with a nondegenerate environment interval, a representation through scalar C^2 submersions u, v and a nonzero induced mixed derivative would imply a fixed left kernel, which contradicts the displayed expression.

One environment activates one object direction; the collection of environments activates both coordinates. The common state (x_1, x_2) is two-dimensional. The example separates pointwise mixed rank from the dimension of a shared relational state. Integrated derivative Gram matrices also register both directions under an environment distribution with interval support, as in active-subspace analysis (Constantine et al. 2014; Zahm et al. 2020). Block projector comparisons assess a proposed common approximation; online appendices B and D give the Gram characterization and inference procedure.

Locally, the quotient admits the expansion $F_0(u, v) = \tilde{u}' A_0 \tilde{v} + o(\|\tilde{u}\| \|\tilde{v}\|)$, where $A_0 = D_u D_v F_0(u_0, v_0)$ is nonsingular. Expenditure and fixed-income valuation already have

exact bilinear primitives. Exact bilinearity yields the projection formulas below, while constant-rank smoothness yields the local information statement.

4. The Target-Relative Measurement Frontier

4.1. Residual variation and local information

For the source and target maps in (2), define the *target-relative residual dimension* and its tangent quotient by

$$(13) \quad d_{O \rightarrow T}(x) = \text{rank}\left(DY_T(x)|_{\ker DY_O(x)}\right),$$

$$(14) \quad \mathcal{Q}_{O \rightarrow T}(x) = \ker DY_O(x) / (\ker DY_O(x) \cap \ker DY_T(x)).$$

The rank counts source-preserving object directions that change target valuations. The term *transport deficiency* refers to this differential dimension. Le Cam deficiency is a separate experiment-approximation distance defined through Markov kernels (Le Cam 1964).

PROPOSITION 2 (Source-fiber sufficiency and information dimension). *Globally, $Y_T = h \circ Y_O$ for a map on attained records exactly when every residual transport set is a singleton, equivalently,*

$$(15) \quad Y_O(x) = Y_O(x') \implies Y_T(x) = Y_T(x').$$

Suppose Y_O and (Y_O, Y_T) have constant ranks near x . Then

$$(16) \quad \dim \mathcal{Q}_{O \rightarrow T}(x) = d_{O \rightarrow T}(x) = \text{rank } D(Y_O, Y_T)(x) - \text{rank } DY_O(x).$$

This is the dimension of the local branch of $\mathcal{R}_{O \rightarrow T}(Y_O(x))$ through $Y_T(x)$. Exact local smooth transport holds precisely when $d_{O \rightarrow T} = 0$. A smooth certificate $C(x) \in \mathbb{R}^k$ and smooth decoder satisfying

$$(17) \quad Y_T(x) = h(Y_O(x), C(x))$$

require $k \geq d_{O \rightarrow T}$. A local certificate of dimension $d_{O \rightarrow T}$ exists in the unrestricted smooth object-side class.

The global equivalence is deterministic sufficiency. The local statement applies the constant-rank theorem to that same source fiber. For a target metric $W_T > 0$, the form

$$(18) \quad g_{O,T,x}([\xi], [\eta]) = DY_T(x)[\xi]' W_T DY_T(x)[\eta]$$

is well defined and positive definite on $\mathcal{Q}_{O \rightarrow T}(x)$. Dimension counts the missing coordinates, while this form measures the target loss attached to them.

The representation in Theorem 2 gives $m(x, z) = \tilde{m}(s_{z_0}(x), z)$, with $\tilde{m}((\alpha, u), z) = \alpha + c(z) + F_0(u, v(z))$ and $c(z) = m(x_0, z) - m(x_0, z_0)$. At a regular local state image $S = s_{z_0}(N_X)$,

$$(19) \quad d_{O \rightarrow T}(x) = \text{rank}\left(D\tilde{Y}_T|_{\ker D\tilde{Y}_O}\right).$$

The derivatives on the right act on $T_S S$. Equation (19) connects the common-state theorem to the record frontier: source-preserving movements and their target consequences can be calculated in the state coordinates. If $z_0 \in O$, every residual state direction has $d\alpha = 0$. The source-fiber criterion also applies directly to any declared measurement map; the quotient supplies reduced coordinates when its conditions hold.

COROLLARY 3 (State-space record bound). *Under the regularity of (19),*

$$(20) \quad d_{O \rightarrow T}(x) \leq \dim \ker D\tilde{Y}_O(s) = \dim S - \text{rank } D\tilde{Y}_O(s) \leq r_M.$$

When the reference environment belongs to O , residual movements lie in the interaction-coordinate tangent space, so $d_{O \rightarrow T}(x) \leq r$. Thus the common-state representation bounds the number of smooth scalar fields that can remain relevant for every declared target family, while (19) gives the target-specific count.

4.2. One residual map, exact and approximate frontiers

In the exact bilinear system, let $q \in \mathbb{R}^n$ be the exposure vector and let $V_O \in \mathbb{R}^{n \times K}$ and $V_T \in \mathbb{R}^{n \times J}$ collect the source and target loadings as columns. Write $\mathcal{V}_O = \text{im } V_O$, $\mathcal{V}_T = \text{im } V_T$, and

$$(21) \quad R_{O,T} = P_{\mathcal{V}_O^\perp} V_T, \quad D(O, T) = \dim(\mathcal{V}_O + \mathcal{V}_T) - \dim \mathcal{V}_O.$$

For $q = q_y + w$ on a source fiber, $w \in \ker V_O'$ and the target innovation is $V_T' w = R_{O,T}' w$. This residual map is the bilinear form of the variation in (3).

To attach average loss, let $\mu_q = \mathbb{E}q$, $\Sigma_q = \text{Var}(q)$ with finite second moments, and $S_q = \text{im } \Sigma_q$. The covariance after best affine prediction from $V_O' q$ is

$$(22) \quad \Sigma_{q|O} = \Sigma_q - \Sigma_q V_O (V_O' \Sigma_q V_O)^\dagger V_O' \Sigma_q.$$

Fix a target metric $W_T > 0$ and define

$$(23) \quad K_{O,T}^{(q)} = \Sigma_{q|O}^{1/2} R_{O,T} W_T R_{O,T}' \Sigma_{q|O}^{1/2}, \quad M_T = V_T W_T V_T'.$$

The exposure distribution and target metric supply the averaging and units of loss. Let $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ denote the eigenvalues of this operator, and write $\tilde{\mathcal{V}}_A = P_{S_q} \mathcal{V}_A$.

THEOREM 3 (Target-relative measurement frontier). *For the exact bilinear measurement system, the residual map $R_{O,T}$ gives the following frontiers.*

- (i) *Exact recovery. On $\mathbb{R}^n$, exact target recovery holds precisely when $R_{O,T} = 0$. The minimum dimension of freely chosen linear certificates is*

$$(24) \quad k_{\text{exact}} = D(O, T) = \text{rank } R_{O,T}.$$

A basis of $\text{im } R_{O,T}$ gives completing exposure projections.

- (ii) *Worst-case loss. For scalar target $t'q$ and exposure ball $\mathcal{U} = B\mathbb{B}_2^n$, let $\text{rad}(A) = \inf_c \sup_{a \in A} |a - c|$. With the supremum over attained source records,*

$$(25) \quad \inf_f \sup_{q \in \mathcal{U}} |t'q - f(V'_O q)| = \sup_y \text{rad } \mathcal{R}_{O \rightarrow t}(y; \mathcal{U}) = B \|P_{\mathcal{V}_O^\perp} t\|_2.$$

- (iii) *Average loss and information repair. With k freely chosen linear certificates and an affine decoder, minimum target-weighted mean-square loss equals*

$$(26) \quad L_k^*(O, T) = \sum_{j>k} \lambda_j, \quad \text{rank } K_{O,T}^{(q)} = \dim(\tilde{\mathcal{V}}_O + \tilde{\mathcal{V}}_T) - \dim \tilde{\mathcal{V}}_O.$$

This rank is the certificate count for zero affine-prediction loss and for exact affine recovery on the exposure envelope $\mu_q + S_q$.

- (iv) *Value of an added record. Put $\Sigma_O = \Sigma_{q|O}$. An added measurement $a'q$ with $a'\Sigma_O a > 0$ changes the residual covariance and target loss by*

$$(27) \quad \Sigma_{O,a} = \Sigma_O - \frac{\Sigma_O a a' \Sigma_O}{a' \Sigma_O a},$$

$$(28) \quad \text{VM}(a \mid O, T) = \frac{a' \Sigma_O M_T \Sigma_O a}{a' \Sigma_O a}.$$

Its value is zero when $a' \Sigma_O a = 0$. A certificate chooses a from the permitted object-side measurements; an environment record restricts a to a feasible loading $v(z)$.

Each clause uses the same residual target map with a different set of declared primitives. Exact rank uses the full exposure region, worst-case loss uses a bounded set, and average loss uses exposure moments together with a target metric. Certificates and environment records share the marginal-value formula in (28); their distinct action sets specify which measurement can be added. In this way, the residual geometry becomes a record-design problem.

Projection and rank identities give the exact count. The average-loss tail uses the Ky Fan variational principle and the geometry of reduced-rank prediction (Fan 1949; Tso 1981). Rank-one covariance updates and target-weighted design connect the marginal value to the optimal-design tradition (Elfving 1952; Kiefer and Wolfowitz 1960; Attia et al. 2018). For Gaussian exposures, the average-loss criterion equals the zero-noise limit of goal-oriented A-optimal design. Appendix C proves the frontier and this correspondence, including singular exposure supports. The exact bilinear assumptions govern these formulas; the nonlinear quotient supplies the local residual dimension through (19).

When target loadings span the centered exposure support, (26) becomes

$$(29) \quad D_{S_q}(O, T) = \text{rank } \Sigma_{q|O} = \dim S_q - \dim \tilde{\mathcal{V}}_O.$$

This is the accounting used in F-MAP: a six-dimensional centered basket space, four independent source valuations, and two remaining target-relevant directions. The dimension statement concerns linear records on the affine exposure envelope. Held-out evaluation checks the trained record and decoder on observed baskets.

At a tolerated average loss ϵ , the effective certificate count is

$$(30) \quad D_{\epsilon, W_T}(O, T) = \min\{k : \sum_{j>k} \lambda_j \leq \epsilon\}.$$

The target metric belongs to the economic loss specification. Under $V_T \mapsto V_T A$, the congruent transformation $W_T \mapsto A^{-1} W_T A^{-\top}$ leaves the residual-loss operator unchanged. Changing the coordinate units while keeping its numerical metric fixed changes the loss being measured.

4.3. Feasible information and institutional capacity

A certificate must be available at the recording date. Let $P(x)$ collect the permitted pre-target object inputs.

PROPOSITION 3 (Admissible information). *The combined input (Y_O, P) permits target recovery precisely when*

$$(31) \quad (Y_O(x), P(x)) = (Y_O(x'), P(x')) \implies Y_T(x) = Y_T(x').$$

Under constant ranks of the combined source and source–target maps, its local smooth counterpart is

$$(32) \quad \ker DY_O \cap \ker DP \subseteq \ker DY_T.$$

Every certificate $C = c \circ P$ enabling smooth local recovery satisfies this necessary kernel inclusion.

When $P(q) = q$, the institution can form a quantity projection when it constructs the record and retain that scalar for future targets. A system with a restricted input P faces the gate in Proposition 3. The unrestricted certificate count in Theorem 3 is the benchmark for a record system that can form the corresponding exposure projections.

Let Z_0 be a finite menu and K an externally supplied capacity. The menu-constrained policy optimum under average loss is

$$(33) \quad L_{\text{menu}}^*(K) = \min_{\substack{O \subseteq Z_0 \\ |O| \leq K}} \text{tr}(M_T \Sigma_{q|O}).$$

The integer K is the institution’s record capacity. The statistical dimension $\hat{r}$ summarizes an estimated spectrum and is a separate calibration diagnostic. The empirical frontier evaluates every supplied value of K and reports $\hat{r}$ alongside it.

The implemented greedy target-directed policy adds the environment maximizing (28). The policy optimum in (33) minimizes the declared calibration objective over all feasible subsets; held-out policy effects evaluate the subsets chosen by the implemented rule. Finite-menu enumeration and one-exchange search measure the search component. With environment costs $c_E(z)$ and object-field costs $c_C(a)$, the institution solves

$$(34) \quad \min_{O,A} \text{tr}(M_T \Sigma_{q|O,A}) + \lambda \left\{ \sum_{z \in O} c_E(z) + \sum_{a \in A} c_C(a) \right\},$$

over its feasible environment and certificate menus, where $\Sigma_{q|O,A}$ is the exposure covariance after projection on both record types. Conditional on the current record, a candidate field enters the one-step solution when its marginal value in (28) exceeds its loss-unit cost λc . The empirical capacity–loss schedule supplies the loss side of this comparison for a common per-field cost; application-specific collection, storage, and compliance costs supply the other side.

For orientation, classical subspace design allows any K -dimensional source span W . With target-direction second moment $\Omega_T = \mathbb{E}[tt']$, the minimum *average squared minimax error* over targets on the exposure ball is

$$(35) \quad \min_{\dim W \leq K} B^2 \mathbb{E} \|P_{W^\perp} t\|_2^2 = B^2 \sum_{j>K} \lambda_j(\Omega_T).$$

This is the leading-eigenspace benchmark. It averages the squared worst-case error for each target; (26) averages prediction error over random exposures. The feasible environment menu in (33) and the declared loss specify the institution's actual choice.

5. Measurement Policy: F-MAP and Treasury

5.1. The F-MAP retention laboratory

USDA Food-at-Home Monthly Area Prices supplies purchase dollars, purchase grams, and unit values for 90 expenditure food product groups in ten metropolitan areas during 2012–2018 (U.S. Department of Agriculture, Economic Research Service 2024). ERS constructs the published series from proprietary Circana scanner inputs and store-level weights. The analysis aggregates dollars and grams into seven consumption groups and forms each group unit value from these totals. Objects are normalized quantity vectors q_i , environments are group price vectors p_e , and the oracle valuation is

$$(36) \quad m(q_i, p_e) = p_e' q_i.$$

Quantities use 100-gram units and sum to 1,000 per basket. The adding-up restriction gives a six-dimensional centered affine support. The categories are dairy, fruit, grains, meat and protein foods, other foods, prepared meals, and vegetables.

The primary retention task uses the first tier of USDA's published food-group hierarchy (U.S. Department of Agriculture, Economic Research Service 2024). Its seven fields describe basket composition in consumption categories: an added scalar can report a weighted shift among fruit, grains, vegetables, and the other named groups. The composition-index dictionary below implements that use. Group unit values incorporate the within-group purchase mix in each environment, so the target is expenditure at that reporting resolution. Product mix and quality composition enter the represented environment together with transaction prices. The normalized system has six centered support directions, allowing the complete capacity and certificate frontier to be evaluated from zero through exact completion. The 34- and 90-group systems retain progressively finer product composition and define their own revaluation maps and capacity comparisons. Their reference capacities cover 8/33 and 13/89 of their centered support dimensions, com-

pared with 4/6 in the primary system; the reported effects therefore pair each resolution with its declared capacity.

Training covers 2012–2015 with 480 area–month environments, validation uses the 120 environments in 2016, and evaluation uses the 240 environments in 2017–2018. Every evaluation basket is valued at every evaluation price vector, producing 57,600 pairs. Training quantities determine exposure moments; validation prices determine source selection and certificate order. Held-out quantities and prices determine evaluation losses.

Capacity is supplied as $K \in \{1, \dots, 6\}$. The four-record point is the reference comparison, and every budget is displayed. Separately, a training-price rule with a 95-percent explained-variation floor and a 0.10 relative eigengap yields $\hat{r} = 4$. This spectral diagnostic is reported alongside the capacity frontier.

For source price matrix P_O , the affine decoder is

$$(37) \quad \hat{m}(q_i, p_t; O) = p'_t \mu_q + p'_t \Sigma_q P'_O (P_O \Sigma_q P'_O)^\dagger P_O (q_i - \mu_q).$$

Its mean-square linear-projection loss is $p'_t \Sigma_{q|O} p_t$. A support-root singular-value decomposition evaluates this decoder numerically. Quantities enter calibration and the oracle, while the decoder receives the K source values and any retained certificates. The design question is which records carry information into later revaluation. The F-MAP series supplies a laboratory for that counterfactual reporting decision; the estimand concerns the retained field menu and downstream decoder.

5.2. The information-policy effect

Fix calibration data S_0 , a capacity K , a decoder class, and a target specification. Policy a chooses a feasible source set O_a of cardinality K . For held-out pair (i, t) , let $\hat{m}_{it}(a)$ use the retained source values and the fitted decoder, and define

$$(38) \quad \ell_{it}(a) = [m(x_i, z_t) - \hat{m}_{it}(a)]^2, \quad \tau_{a:b} = \frac{1}{N_x N_t} \sum_{i,t} [\ell_{it}(b) - \ell_{it}(a)].$$

The oracle value $m(x_i, z_t)$ and the trained policy maps give both losses for every evaluation pair. Thus $\tau_{a:b}$ is the observed finite-population effect of assigning one retained-information rule in place of the other. The object, target environment, calibration data, capacity, and decoder class are held fixed. This intervention concerns the information supplied to a future revaluation task.

For random source selection A , the comparator is $\mathbb{E}_A \ell_{it}(A)$. Uniform draws estimate this policy expectation. The random-policy tail reports the fraction of draws attaining loss weakly below the greedy target-directed rule, with the finite-draw correction. Repeated-population references condition on the trained rules. Two-way clustering treats object

and target dimensions separately, a node jackknife removes each area-month from both roles, and the forward cell assigns distinct calendar years to the two roles. The finite evaluation-array effect is computed directly.

For simultaneous evaluation, index the 240 held-out area-month nodes by g and let $\widehat{\theta}_{(-g)}$ delete node g from both the basket and target dimensions. The pseudo-value $\xi_g = G\widehat{\theta} - (G-1)\widehat{\theta}_{(-g)}$ supplies a node-level linearization. One common set of 4,999 Gaussian multipliers acts on the pseudo-values for exchange-policy loss and its effect against dispersed acquisition at capacities one through five. The 95-percent critical value is the 0.95 quantile of the maximum studentized perturbation. Capacity six is reported as algebraic closure. A confidence-protected capacity is the smallest K whose simultaneous RMSE upper bound lies below a supplied basis-point threshold.

Published unit values estimate the environment loadings. Let $\widehat{P}$ denote the published loading array and let $\widehat{S}$ contain its public standard errors. The loading sensitivity calculation draws $P^{(b)} = \widehat{P} + \widehat{S} \odot E^{(b)}$, propagates EFPG uncertainty to the seven groups with fixed purchase-gram weights, refits each source policy, and evaluates both the published and re-extracted target arrays. Each draw runs exchange search from the draw-specific greedy set and from the published exchange set, retaining the lower calibration loss. Source overlap records identity stability; principal angles and projector distance record stability of the retained information span; the draw-specific excess calibration loss of the published set records policy regret. Published F-MAP fields supply marginal standard errors. The baseline calculation uses a diagonal covariance specification, and a separate equicorrelation grid varies cross-EFPG dependence. A second reconstruction uses the public unweighted dollar and gram fields. These exercises address loading estimation and store weighting; the finite-array estimand in (38) remains tied to its declared valuation array.

The selection-aware calculation crosses each of the 499 loading re-extractions and its reselected exchange and dispersed policies with ten node-multiplier perturbations of the corresponding held-out loss matrices. This produces 4,990 draws that propagate discrete policy reselection under the published diagonal unit-value covariance and repeated-node evaluation uncertainty. The resulting percentiles form a joint sensitivity range for these two declared uncertainty sources. The equicorrelation grid remains a separate covariance sensitivity analysis because F-MAP publishes marginal unit-value standard errors without their full covariance matrix.

5.3. The capacity frontier and environment choice

At each capacity, greedy target-directed acquisition adds the training environment giving the largest reduction in validation target loss. Dispersed acquisition shares the first environment and then selects the farthest point in standardized training-price coordinates. Uni-

TABLE 1. Held-out loss at each record capacity

Budget K	Greedy RMSE	Dispersed RMSE	Random RMSE	Certificates $6 - K$
1	0.679937	0.679937	1.045165	5
2	0.419336	0.649119	0.645690	4
3	0.260297	0.583402	0.439927	3
4	0.153629	0.288628	0.293871	2
5	0.075742	0.153754	0.175604	1
6	2.0×10^{-12}	1.5×10^{-13}	4.2×10^{-12}	0

Each capacity is supplied independently of the training-price dimension diagnostic. The random column is the square root of expected MSE across 4,999 uniform source sets at that capacity. The last column records the completing certificate count for the greedy and dispersed source rules. Every row uses the same calendar split and evaluation array.

TABLE 2. Greedy-policy effects at the four-record reference capacity

Comparator	Comparator MSE	Greedy MSE	MSE reduction	95% effect interval
Dispersed	0.083306	0.023602	71.7%	[0.040935, 0.078474]
Uniform random	0.086360	0.023602	72.7%	[0.052767, 0.072749]

Effects are comparator loss minus implemented greedy target-directed loss over 57,600 pairs. Equation (33) defines the calibration menu optimum; appendix E reports finite-menu gaps for the implemented search rules. Intervals use two-way clustering by basket and target environment conditional on the trained policies. The random row uses Monte Carlo expected loss.

form acquisition draws K distinct training environments. The greedy target-directed rule implements the marginal value in (28) on the feasible price menu. The menu-constrained policy optimum in (33) is the calibration benchmark; the tables evaluate the implemented greedy policy on held-out values.

At $K = 5$, greedy and dispersed RMSE are 0.0757 and 0.1538. At $K = 6$, source values span the centered basket support and the deterministic rules recover the target array to numerical precision. The record frontier and the certificate frontier reach the same six-coordinate envelope through environment loadings or retained quantity projections.

Of 4,999 draws, 92 attain loss weakly below greedy loss, giving the corrected tail $(92+1)/5,000 = 0.0186$. The complete-array effect is an observed finite-population contrast. A delete-one-node jackknife removes each area-month from both basket and target roles and gives [0.03901, 0.08040]. A forward cell formed from 2017 baskets and 2018 target prices gives an effect of 0.04999, corresponding to a 59.9-percent MSE reduction, with a two-way clustered interval [0.02708, 0.07289]. Appendix E reports the node and forward-calendar calculations.

The four-record exchange policy reduces RMSE from 0.2886 under dispersed acquisition to 0.1364. The difference is \$0.1523 per normalized 100-kilogram basket, equal to 4.612

TABLE 3. Simultaneous inference and confidence-protected capacity

K	Exchange MSE	RMSE (bp)	Effect	Simultaneous 95% interval	UCB RMSE (bp)
1	0.462314	20.60	0.00000	[0.00000, 0.00000]	22.77
2	0.157026	12.00	0.26433	[0.17685, 0.35181]	13.38
3	0.067946	7.90	0.27241	[0.17894, 0.36589]	9.02
4	0.018594	4.13	0.06471	[0.03668, 0.09274]	4.74
5	0.004691	2.07	0.01895	[0.01165, 0.02624]	2.48
6	0.000000	0.00	0.00000	[0.00000, 0.00000]	0.00

Effects are dispersed loss minus exchange-policy loss. A common 4,999-draw Gaussian multiplier acts on delete-one-area-month pseudovalues for the effect and exchange MSE at capacities one through five. The displayed intervals and MSE upper bounds use the resulting maximum studentized critical value of 2.676. Capacity six is exact algebraic closure. RMSE is measured in basis points of the \$330.145 calibration benchmark expenditure.

basis points of the \$330.145 calibration benchmark expenditure. MSE falls by 0.064712 squared dollars per basket–target pair. For N future revaluations, a cost-weighted loss criterion admits incremental search and implementation cost up to $N\kappa(0.064712)$, where κ converts squared-dollar loss into the institution’s cost unit. Both rules retain four scalar fields, so the common per-field storage component cancels in this comparison.

At four records, the fixed-policy effect is 0.06471 with simultaneous interval [0.03668, 0.09274]. Exchange RMSE is 4.13 basis points and its simultaneous upper bound is 4.74. The smallest protected capacities are three records for a 10-basis-point ceiling, four for 5 basis points, five for 2.5 basis points, and six for 1 basis point. These statements condition on the validation-selected policies before the 2017–2018 evaluation.

The loading sensitivity analysis uses the public standard error attached to each weighted EFPG unit value. Crossing 499 draw-specific policy reselections with ten held-out node perturbations gives 4,990 draws, a 75.9-percent median MSE reduction, and a 44.2–90.1 percentile range; all 4,990 loss effects are positive. Across five equicorrelation specifications for the unavailable EFPG covariance, the exchange median ranges from 72.8 to 76.1 percent and every 2.5th percentile exceeds 31 percent. Appendix E gives the selection-aware construction and its covariance scope.

Finite-menu optimization. The finite-menu check enumerates every one- and two-record subset of the full 480-environment training menu and every subset of size one through four in 12 reduced menus. Each reduced menu draws two training environments from each metropolitan area. In total, the comparison evaluates 189,780 subsets using training exposure moments and validation targets. At $K = 1$, greedy attains the menu optimum. At $K = 2$ in the full menu, its validation loss is 8.65 percent above the optimum; one-for-one exchange from the greedy set reaches that optimum and lowers held-out MSE from

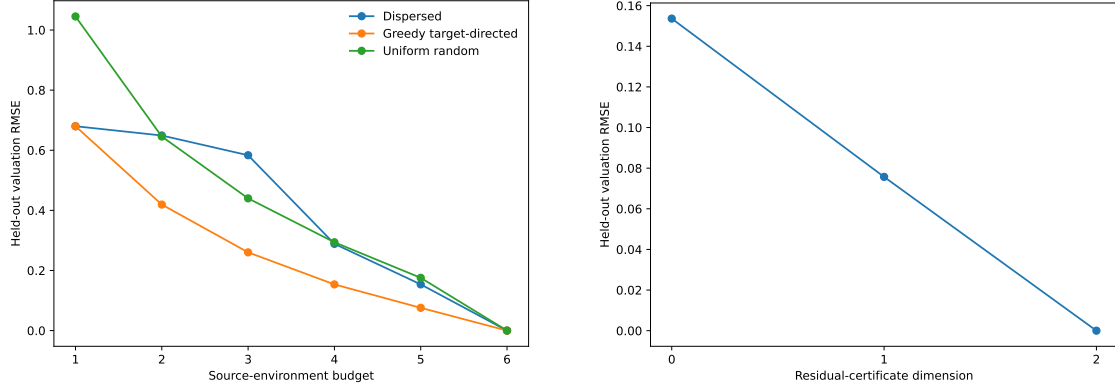


FIGURE 1. The two measurement margins. The left panel varies retained source capacity from one to six environments. The random curve is the square root of expected MSE across uniform source-set draws. The right panel adds calibration-based certificates to four greedy source values. Both panels evaluate the same 57,600 held-out revaluations.

0.175843 to 0.157026. Across the reduced menus, greedy mean validation gaps are 11.74, 31.11, and 41.17 percent at capacities two, three, and four; exchange search reduces them to 0.00, 0.19, and 3.08 percent. On the full menu at $K = 4$, six accepted exchanges lower calibration MSE from 0.016848 to 0.014863 and held-out MSE from 0.023602 to 0.018594, a 77.7-percent reduction relative to dispersed acquisition. The disjoint 2017-to-2018 evaluation gives 69.9 percent and an effect interval $[0.03599, 0.08060]$. One-exchange search is a Fedorov-style local routine; experimental-design theory supplies guarantees under criterion and conditioning assumptions (Lau and Zhou 2022). Appendix E reports the search paths, maximum gaps, and held-out comparisons. These calculations keep the record-information criterion separate from the combinatorial routine used to search its menu.

5.4. Two residual dimensions and two completing certificates

At $K = 4$, greedy source valuations have rank four on the six-dimensional centered basket support. Evaluation target prices span all six support directions. Equation (29) therefore predicts two target-relevant missing coordinates. The residual covariance has rank two. The first two certificates are formed from the validation target-loss operator on this calibration residual space.

One certificate removes 75.7 percent of remaining MSE; two recover every held-out target value to numerical precision. The first certificate’s paired loss reduction has a two-way clustered 95-percent interval of $[0.01323, 0.02251]$. These certificates record quantity-composition directions that the four source expenditure numbers leave unresolved. The affine-support identity predicts their count; the evaluation checks the trained directions and decoder on later baskets and prices.

TABLE 4. Residual information and held-out revaluation

Retained information	Certificates	RMSE	MSE	MSE removed
Four source values	0	0.153629	0.023602	—
First residual certificate	1	0.075711	0.005732	75.7%
Two residual certificates	2	5.1×10^{-14}	2.6×10^{-27}	100.0%

The centered basket support has dimension six, the source-loading rank is four, and the target-loading rank is six. Certificate directions use training quantities and validation prices. The table evaluates all 57,600 held-out pairs. Numerical recovery is assessed against an absolute value-error tolerance of 10^{-9} .

Each certificate can be stored as one composition index. Let $s_{ig} = q_{ig}/1,000$ be group g 's share of basket mass. The certificate directions are recoded as

$$(39) \quad C_j(q_i) = 100 \sum_{g=1}^7 w_{jg} s_{ig}, \quad \sum_{g=1}^7 w_{jg} = 0, \quad \max_g |w_{jg}| = 1.$$

These scalar fields use a common weight dictionary. Moving one percentage point of basket mass from vegetables to fruit changes the first index by 1.694 points; moving that share from vegetables to grains changes the second by 1.519 points. This affine recoding preserves the information in the certificates. Appendix E supplies the seven-group weights and the retained-field construction. Revaluation from four expenditure fields and two composition-index fields closes the two-dimensional deficit to numerical precision.

The count is reproduced across the capacity frontier. For both greedy and dispersed source sets, the completing certificate count equals $6 - K$ at every $K = 1, \dots, 6$. The counts agree at relative rank tolerances 10^{-8} , 10^{-10} , and 10^{-12} . Ten leave-one-metropolitan-area-out calibrations repeat the four-record exercise: training and validation exclude one area, and evaluation uses that area's 24 later baskets against all 240 later targets. Each calibration again requires two completing certificates. Appendix E reports the closure errors and calibration-sample comparisons.

5.5. The marginal value of retained capacity

Let $L_{\text{eval}}(K)$ be held-out MSE along the greedy acquisition sequence. At $K = 0$, each target is valued using the training mean basket. One additional retained scalar saves

$$(40) \quad \Delta L_{\text{eval}}(K) = L_{\text{eval}}(K-1) - L_{\text{eval}}(K), \quad K = 1, \dots, 6.$$

The calibration analogue equals the marginal record value in (28). Equation (40) evaluates that increment on subsequent baskets and targets.

The fourth record saves 0.044153 in squared-dollar loss per basket-target pair, or 65.2 percent of the three-record loss. The fifth saves 0.017865, or 75.7 percent of the four-record

TABLE 5. Marginal value of one retained scalar

K	MSE	Saved per scalar	Previous loss saved (%)	RMSE (basis points)
0	40.889540	—	—	193.687
1	0.462314	40.427227	98.9	20.595
2	0.175843	0.286471	62.0	12.702
3	0.067755	0.108088	61.5	7.884
4	0.023602	0.044153	65.2	4.653
5	0.005737	0.017865	75.7	2.294
6	4.0×10^{-24}	0.005737	100.0	6.1×10^{-11}

MSE and saved loss are in squared dollars for the normalized 100-kilogram basket. Each transition adds one source-value field. The percentage column divides saved loss by the preceding capacity’s MSE. The last column is $10,000\sqrt{L_{\text{eval}}(K)}/\bar{v}_{\text{cal}}$, where $\bar{v}_{\text{cal}} = \bar{p}'_{\text{validation}}\mu_q = \330.145 is the calibration benchmark expenditure. One basis point is 10^{-4} of that benchmark.

loss. At four records, RMSE equals 4.653 basis points of benchmark expenditure; at five it equals 2.294. The sixth record completes the centered exposure support. This schedule expresses capacity in retained scalar fields and accuracy in valuation units. An institution can compare its entries with a supplied record cost and a chosen valuation-loss criterion.

5.6. Treasury exposure records

Treasury valuation applies the same calculation to dated promised cash flows c_i and discount curves $d(z_t)$:

$$(41) \quad m(c_i, z_t) = d(z_t)'c_i.$$

Federal Reserve curves determine the discount coordinates, and Treasury auction records determine the cash-flow schedules. The reference experiment retains one source valuation. Duration, duration plus convexity, and five key-rate measurements add cash-flow projections with exposure to target discount directions. These fields give economic names to the object-side certificates: each retained exposure summary supplies a scalar loading on the promised payment schedule.

The validation-optimal residual certificate minimizes the calibration loss over one additional linear field. Its held-out RMSE is 2.234, while the duration field attains 0.789. Optimality refers to the training-moment and validation-target criterion; the later cash-flow and discount populations determine the displayed ordering.

The Treasury records illustrate the approximate part of Theorem 3: each certificate removes target-weighted residual variation, with losses assessed in valuation units. Calendar-regime comparisons reject equality of the leading rank-one discount space with finite-draw $p = 0.0002$. This rotation evaluates the stability of a one-direction approximation. Certifi-

TABLE 6. Treasury certificate repair

Additional information	Held-out RMSE	MSE removed
Source valuation alone	14.563	—
Duration	0.789	99.71%
Duration and convexity	0.631	99.81%
Five key-rate certificates	0.212	99.98%
Validation-optimal residual certificate	2.234	97.65%

RMSE is in dollars per \$100 face value. Reductions are relative to the one-source valuation decoder. Validation optimality uses 1990–2007 cash-flow moments and 2008–2015 target curves. Held-out losses use 2016–2025 cash flows and curves. Appendix F gives the construction and complete comparisons.

cate value is calculated against the actual target discount family, whose residual directions determine the repair task. The exact bilinear cash-flow map and its target-relative frontier remain the valuation primitives.

Active-space inference supports this interpretation of a proposed common approximation. It is reported in appendices B and D together with the eigengap gate and bootstrap conditions. Treasury robustness and auxiliary scanner diagnostics appear in appendix F; their role is validation of the residual-information calculation across environments and valuation domains.

6. Conclusion

Economic measurement carries the same object across evaluation environments. Stable mixed kernels yield relational coordinates whose dimension matches the interaction rank, and a reference value supplies the level information. Source records restrict this state to a fiber. The target image of that fiber determines whether recovery is exact and how much additional smooth information is required locally. In exact bilinear systems, the same residual target map determines bounded-set error, average affine-prediction loss, and the marginal value of another record. Certificates and environment valuations apply this calculation to different feasible menus of fields.

In F-MAP, four target-directed source records reduce held-out MSE by 71.7 percent relative to dispersed selection. One-exchange refinement raises the reduction to 77.7 percent. Its fixed-policy effect has a simultaneous interval of $[0.0367, 0.0927]$, and combining public-unit-value policy reselection with held-out node perturbations gives a 75.9-percent median reduction and a $[44.2, 90.1]$ central range. The confidence-protected capacity is four fields for a 5-basis-point RMSE ceiling under the declared evaluation design. Two calibrated composition indices complete the two residual directions and recover later valuations to numerical precision. Treasury duration, convexity, and key-rate exposures apply the same record-completion calculation to discount environments. A downstream

welfare calculation would also require a decision loss and the costs of collecting and retaining records. To use the frontier, an institution must therefore specify its future targets, feasible records, capacity, and costs.

References

- Ingvild Almås, Orazio Attanasio, and Pamela Jervis. Presidential address: Economics and measurement: New measures to model decision making. *Econometrica*, 92(4):947–978, 2024. doi: 10.3982/ECTA21528.
- Ahmed Attia, Alen Alexanderian, and Arvind K. Saibaba. Goal-oriented optimal design of experiments for large-scale bayesian linear inverse problems. *Inverse Problems*, 34(9):095009, 2018. doi: 10.1088/1361-6420/aad210.
- David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.
- Qiao Chen, Élise Arnaud, Ricardo Baptista, and Olivier Zahm. Coupled input–output dimension reduction: Application to goal-oriented bayesian experimental design and global sensitivity analysis. *SIAM Journal on Scientific Computing*, 47(5):A2403–A2430, 2025. doi: 10.1137/24M1670421.
- Pierre-André Chiappori and Ivar Ekeland. Aggregation and market demand: An exterior differential calculus viewpoint. *Econometrica*, 67(6):1435–1457, 1999. doi: 10.1111/1468-0262.00085.
- Paul G. Constantine, Eric Dow, and Qiqi Wang. Active subspace methods in theory and practice: Applications to kriging surfaces. *SIAM Journal on Scientific Computing*, 36(4):A1500–A1524, 2014. doi: 10.1137/130916138.
- W. Erwin Diewert. Exact and superlative index numbers. *Journal of Econometrics*, 4(2):115–145, 1976. doi: 10.1016/0304-4076(76)90009-9.
- Gustav Elfving. Optimum allocation in linear regression theory. *The Annals of Mathematical Statistics*, 23(2):255–262, 1952. doi: 10.1214/aoms/1177729442.
- Ky Fan. On a theorem of Weyl concerning eigenvalues of linear transformations I. *Proceedings of the National Academy of Sciences*, 35(11):652–655, 1949. doi: 10.1073/pnas.35.11.652.
- Steven M. Goldman and Hirofumi Uzawa. A note on separability in demand analysis. *Econometrica*, 32(3):387–398, 1964. doi: 10.2307/1913043.
- W. M. Gorman. Separable utility and aggregation. *Econometrica*, 27(3):469–481, 1959. doi: 10.2307/1909472.
- W. M. Gorman. The structure of utility functions. *Review of Economic Studies*, 35(4):367–390, 1968. doi: 10.2307/2296766.
- W. M. Gorman. Notes on Divisia indices. In Charles Blackorby and Anthony F. Shorrocks, editors, *Separability and Aggregation: The Collected Works of W. M. Gorman, Volume I*, pages 49–60. Oxford University Press, 1996. doi: 10.1093/0198285213.003.0004. Originally circulated as a 1970 typescript.
- H. A. John Green. *Aggregation in Economic Analysis: An Introductory Survey*. Princeton University Press, Princeton, NJ, 1964.
- Stefan Hoderlein and Arthur Lewbel. Regressor dimension reduction with economic constraints: The example of demand systems with many goods. *Econometric Theory*, 28(5):1087–1120, 2012. doi: 10.1017/S0266466612000412.

- J. Kiefer and J. Wolfowitz. The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366, 1960. doi: 10.4153/CJM-1960-030-4.
- Lap Chi Lau and Hong Zhou. A local search framework for experimental design. *SIAM Journal on Computing*, 51(4):900–951, 2022. doi: 10.1137/20M1386542.
- Lucien Le Cam. Sufficiency and approximate sufficiency. *The Annals of Mathematical Statistics*, 35(4):1419–1455, 1964. doi: 10.1214/AOMS/1177700372.
- John M. Lee. *Introduction to Smooth Manifolds*. Springer, New York, 2 edition, 2013.
- Kuang-Yao Lee, Bing Li, and Francesca Chiaromonte. A general theory for nonlinear sufficient dimension reduction: Formulation and estimation. *Annals of Statistics*, 41(1):221–249, 2013. doi: 10.1214/12-AOS1071.
- Wassily Leontief. A note on the interrelation of subsets of independent variables of a continuous function with continuous first derivatives. *Bulletin of the American Mathematical Society*, 53(4):343–350, 1947a. doi: 10.1090/S0002-9904-1947-08796-6.
- Wassily Leontief. Introduction to a theory of the internal structure of functional relationships. *Econometrica*, 15(4):361–373, 1947b. doi: 10.2307/1905335.
- André Nataf. Sur la possibilité de construction de certains macromodèles. *Econometrica*, 16(3):232–244, 1948. doi: 10.2307/1907277.
- Susanne M. Schennach. Measurement systems. *Journal of Economic Literature*, 60(4):1223–1263, 2022. doi: 10.1257/jel.20211355.
- Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proceedings of the 37th Annual Allerton Conference on Communication, Control, and Computing*, pages 368–377, 1999. URL <https://arxiv.org/abs/physics/0004057>.
- M. K.-S. Tso. Reduced-rank regression and canonical analysis. *Journal of the Royal Statistical Society: Series B (Methodological)*, 43(2):183–189, 1981. doi: 10.1111/j.2517-6161.1981.tb01169.x.
- U.S. Department of Agriculture, Economic Research Service. Food-at-home monthly area prices, 2012–2018, 2024. URL <https://www.ers.usda.gov/data-products/food-at-home-monthly-area-prices/documentation>. Data product; documentation updated April 2026 and accessed August 2026.
- Keyi Wu, Peng Chen, and Omar Ghattas. A fast and scalable computational framework for large-scale high-dimensional bayesian optimal experimental design. *SIAM/ASA Journal on Uncertainty Quantification*, 11(1):235–261, 2023. doi: 10.1137/21M1466499.
- Olivier Zahm, Paul G. Constantine, Clémentine Prieur, and Youssef M. Marzouk. Gradient-based dimension reduction of multivariate vector-valued functions. *SIAM Journal on Scientific Computing*, 42(1):A534–A558, 2020. doi: 10.1137/18M1221837.
- Shijie Zhong, Wanggang Shen, Tommie Catanach, and Xun Huan. Goal-oriented bayesian optimal experimental design for nonlinear models using markov chain monte carlo. *SIAM/ASA Journal on Uncertainty Quantification*, 14(1):19–47, 2026. doi: 10.1137/24M1649344.